

Hands-On-Learning: Responsible AI Risk Reflection

Learning Objectives

By the end of this hands-on practice, you will be able to:

- Identify ethical and societal risks in an AI credit-scoring context
- Match risks to relevant Responsible AI principles (Fairness, Transparency, Accountability, Privacy)
- Document realistic mitigation strategies using a structured template
- Reflect on how proactive risk identification builds trust and accountability



Description

In this hands-on Visual Studio Code practice, you will step into the role of a Responsible AI analyst at FinWise Credit, a fintech startup building an AI-powered loan recommendation model. Your task is to identify ethical and societal risks in automated credit decisions and map them to core Responsible AI principles such as fairness, transparency, and accountability. You will complete a structured Responsible AI Risk Table and propose mitigations that reduce harm and build public trust.

This hands-on practice helps you connect abstract Responsible AI values to real design decisions. Working with a simple JSON risk-tracking file, you will document 3–5 potential risks—such as data bias, lack of explainability, or unfair denials—and suggest corresponding mitigations. By codifying these insights, you'll practice early-stage ethical evaluation that can be reused throughout an AI system's lifecycle.

Structure of the Activity

Dataset Information

Filename: risk_table.json

Format: JSON template with fields for Risk ID, description, Responsible AI principle, and mitigation notes

Key Features: Empty template structure for documenting risks; supports Fairness, Transparency, Accountability, and Privacy principles

Tools Used:

- Visual Studio Code (any recent version)
- Python 3.10 or higher
- No external libraries required (uses built-in JSON module)

Setup Requirements

- **Technical Prerequisites:** Python 3.10+ installed and accessible from the command line
- **System Requirements:** Any operating system with Visual Studio Code support
- **Required Files in Working Directory:** risk_table.json (JSON template), reflection.md (reflection template)

Scenario

You are a Responsible AI analyst at FinWise Credit, a fintech startup preparing to deploy an AI-powered loan recommendation model within the next quarter. No formal ethics review has been conducted yet. Leadership has asked you to identify potential ethical and societal risks before deployment and document mitigation strategies that will help the company build public trust while reducing harm to loan applicants.

Objective

Complete a Responsible AI Risk Table with at least 3 documented risks, each mapped to a core RAI principle with a proposed mitigation strategy, and write a reflection on early ethics integration.



Getting Started

- 1 Open Visual Studio Code and create a new folder for your hands-on practice work
- 2 Create a new file named `risk_table.json` with an empty JSON array structure
- 3 Create a new file named `reflection.md` for your written reflection
- 4 Verify Python is installed by running `python --version` in the terminal

Instructions

Activity 1: Identify and Record Ethical Risks

Objective

Identify at least 3 ethical or societal risks in the AI loan recommendation system and document them in a structured JSON format.



Input/Data

Your risk_table.json file (initially empty)

Context: FinWise Credit's AI loan model uses historical lending data to make loan approval recommendations

Steps to Complete

Step 1: Brainstorm potential ethical risks by considering questions such as: What biases might exist in historical lending data? How might the model affect different demographic groups? Can applicants understand why they were approved or denied?

Step 2: Design a JSON structure that captures: a unique risk ID, a short description of the risk, the associated Responsible AI principle (Fairness, Transparency, Accountability, or Privacy), and a mitigation note

Step 3: Document at least 3 distinct risks in your JSON file, ensuring each risk is mapped to the most appropriate Responsible AI principle

Step 4: Write a one-sentence mitigation proposal for each risk that addresses the core concern

Expected Output

A valid JSON file containing an array of risk objects

Each risk object should include: risk_id, description, principle, and mitigation fields

At least 3 risks covering different aspects of the AI system

Tips

Consider risks across multiple categories: data quality, model behavior, user experience, and organizational processes

Use consistent naming conventions for your JSON keys (e.g., snake_case)

Validate your JSON syntax by opening the file in VS Code—it will highlight errors

Activity 2: Review and Validate Risk Documentation

Objective

Create a Python script to read and display your risk data in a formatted table, then review and refine your entries for clarity.



Input/Data

Your completed risk_table.json from Activity 1

Python's built-in json module (no installation required)

Steps to Complete

Step 1: Create a Python script that loads your risk_table.json file using the json module

Step 2: Implement a function that formats and prints each risk entry in a readable table format, displaying all fields (ID, description, principle, mitigation)

Step 3: Run your script and review the output for completeness and clarity

Step 4: Revise any entries in your JSON file to ensure descriptions are concise, non-redundant, and use appropriate Responsible AI terminology

Expected Output

A Python script that successfully reads and displays JSON data

Formatted console output showing all risks in a readable table

Refined risk entries with clear, professional language

Tips

Use Python string formatting to create aligned columns in your output

Handle potential errors gracefully (e.g., file not found, invalid JSON)

Keep mitigation statements action-oriented and specific

Activity 3: Reflect on Early Ethics Integration

Objective

Write a reflection on the importance of evaluating ethics early in the AI development cycle.



Input/Data

Your reflection.md file

Insights from Activities 1 and 2

Steps to Complete

Step 1: Open your reflection.md file and add a heading for your reflection

Step 2: Write approximately 200 words explaining how evaluating ethics early in the design cycle can reduce downstream harm

Step 3: Include specific examples from the FinWise Credit scenario to illustrate your points

Step 4: Discuss how proactive risk identification can strengthen user confidence and organizational accountability

Expected Output

A markdown file with approximately 200 words of reflective writing

Clear connection between early ethics evaluation and harm reduction

Specific references to the loan recommendation scenario

Tips

Consider the costs of fixing ethical issues after deployment versus addressing them during design

Think about both direct harms (to individuals) and indirect harms (to organizational reputation)

Reference specific risks you identified to ground your reflection in concrete examples

Compile Your Findings

Requirement 1: Responsible AI Risk Table

Completed risk_table.json file

Include:

- At least 3 documented risks with unique IDs
- Each risk mapped to appropriate Responsible AI principle
- One-sentence mitigation strategy for each risk
- Valid JSON syntax

Requirement 2: Risk Analysis Script

Python script that reads and displays risk data

Include:

- Functional JSON file loading
- Formatted table output
- Error handling for common issues

Requirement 3: Ethics Reflection

Completed reflection.md file

Include:

- Approximately 200 words of reflective analysis
- Connection between early ethics evaluation and reduced downstream harm
- Discussion of trust and accountability implications

Summary

This activity teaches ethical risk identification and documentation through practical implementation in a realistic fintech scenario. You've learned how to systematically identify Responsible AI risks, map them to core principles, and document mitigation strategies—skills directly applicable in AI governance, product development, and compliance roles across any industry deploying AI systems.

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.